

Checkpoint — Anomaly Scenario Redesign & Feature Pipeline Overhaul

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1 Context and Motivation

Both anomaly detection models (Autoencoder and LSTM) trained on the original enriched dataset produced near-random performance:

Model	Test AUC
Autoencoder (reconstruction MSE)	0.577
LSTM (8 prediction heads)	0.564

Root cause analysis identified two independent problems:

1. **Generator gap:** Of the original 13 anomaly scenarios, only 2 (amount_spike, round_amount) produced detectable changes in the per-event feature space. The remaining scenarios were either unimplemented stubs or modified attributes invisible to the model (e.g., timing_anomaly shifting hours that were never captured as features).
2. **Feature design flaw:** The enriched event contained ~74 features, of which ~60 were raw profile fields (rolling statistics, JS-divergence scores, peer z-scores) copied directly from weekly profile snapshots. These fields were *identical* across all events from the same client within the same snapshot week, drowning out the ~10 per-event features that actually varied.

The decision was made to pursue **Path A**: redesign the anomaly generator and feature pipeline so that every scenario produces an observable footprint, then regenerate and retrain.

2 Anomaly Scenario Audit and Redesign

2.1 Audit Outcome

Each of the original 13 scenarios was evaluated against the question: “*When this anomaly fires on a single event, which enriched features actually change in a way the model can distinguish from normal?*”

Scenario	Status	Reason
amount_spike	Detectable	z_amount, amount_to_mean_ratio fire directly

Scenario	Status	Reason
round_amount	Detectable	Amount features fire; missing is_round_amount feature
duplicate_virement	Not implemented	Listed in plan but no code in generator
volume_spike (emp.)	No mechanism	No employee-level injection system
frequency_burst	Stub (pass)	No event increase; profile-based counts won't shift
cumulative_threshold	Not implemented	No code in generator
timing_anomaly	Weak signal	Hours 17–18 borderline; feature existed but signal too subtle
client_concentration	No mechanism	Employee-level injection missing
smurfing	Per-event only	Single event in 8k–10k band not suspicious alone
cheque_structuring	Not implemented	—
money_mule	Not implemented	—
behavioral_drift	Profile-level	JS-divergence doesn't change per event
cross_actor	Not implemented	—

2.2 Redesigned Scenario Set (9 Scenarios)

Five scenarios were cut (behavioral_drift, volume_spike, client_concentration, money_mule, cross_actor) due to scope constraints or fundamental incompatibility with per-event detection. One employee scenario was added (repeated_client_employee) to preserve the dual-actor model.

Scenario	Difficulty	Injection Logic	Duration
amount_spike	Easy	Multiply one event's amount by 5–10×	1 day
round_amount	Easy	Force amount to 5k/10k/15k/20k DT	1 day
duplicate_virement	Easy	Clone a virement (same amount, beneficiary, ±5–30 min)	1 day
timing_anomaly	Medium	Shift all events to hours 5–7 or 19–21	3 days
frequency_burst	Medium	Generate 5–8× normal daily events	3 days
cumulative_threshold	Medium	3–5 versements summing >10,000 DT, each <9,500 DT	1 day
smurfing	Hard	1–2 versements/day in 8,000–9,800 DT band	7 days
cheque_structuring	Hard	1–2 cheques/day in 25,000–29,900 DT range	7 days
repeated_client_employee	Hard	5–8 events with forced same employee_id	1 day

Key architectural change: anomalous events are generated *alongside* normal events (additive injection) rather than modifying them in place. The anomaly window check occurs *before* the daily probability check, ensuring anomalous events always fire during their assigned window.

3 Feature Pipeline Overhaul

3.1 Problem: Profile Dump

The original enriched event concatenated ~ 194 client profile fields with ~ 10 per-event fields. Since profiles are computed from weekly snapshots, all events from the same client within the same week shared identical profile values. The model input was dominated by static features that provided no discrimination between normal and anomalous events from the same client.

3.2 Solution: Contrast Features

The enriched event was redesigned around three categories, totaling **30 features**:

1. **Raw event features** (7 features): amount, hour, operation_type (4 one-hot), is_home_branch. These vary per event.
2. **Contrast features** (17 features): Each compares the current event against the client baseline. Includes z_amount, amount_to_mean_ratio, threshold flags (is_above_threshold, is_near_threshold, is_round_amount, is_near_cheque_ceiling), op_type_probability, timing contrasts (is_outside_hours, day_distance_from_preferred), counterparty contrast (is_new_counterparty), and buffer-derived velocity features (tx_count_24h, tx_count_7d, cumulative_amount_24h, has_duplicate_recent, near_threshold_count_7d).
3. **Context features** (6 features): client_archetype (5 one-hot), account_age_days. Static but useful for the model to calibrate expectations by client type.

3.3 Oracle Profile Shortcut

The weekly profile snapshot pipeline was replaced by exporting client personality parameters (amount_mean, amount_std, operation_mix, preferred_day, home_branch) directly from the generator's Client objects. This serves as the baseline for contrast feature computation and eliminates the profile snapshot $\rightarrow$ as-of join pipeline, removing the data leakage risk identified in earlier phases.

In production, this baseline would be estimated from historical transaction data. The oracle approach is noted as a simplification that provides an upper bound on model performance.

3.4 Buffer-Based Velocity Features

Velocity features (tx_count_24h, cumulative_amount_24h, etc.) are computed from a per-client rolling buffer of the last 50 events, maintained during enrichment. Two per-employee buffer features are also computed: employee_tx_count_24h and same_employee_client_count_24h. The current event is added to the buffer *after* feature computation to prevent self-reference.

4 Regenerated Dataset Statistics

Metric	Value
Total events	244,141
Unique clients	7,382
Simulation period	180 days
Feature count	30
Anomaly events	1,580
Anomaly rate	0.647%

Scenario	Count	Difficulty
frequency_burst	509	medium
smurfing	302	hard
cheque_structuring	260	hard
repeated_client_employee	181	hard
timing_anomaly	98	medium
cumulative_threshold	77	medium
duplicate_virement	70	easy
amount_spike	43	easy
round_amount	40	easy

Feature separation analysis confirmed that every contrast feature shows meaningful divergence between normal and anomalous events (e.g., `z_amount`: normal mean ≈ 0.0 vs anomaly mean ≈ 308 ; `same_employee_client_count_24h`: ratio $\approx 82\times$).

5 Model Results

5.1 Data Split

Chronological split with 21-day warm-up filter (buffer population period):

Split	Day Range	Events	Anomalies
Train (normal only)	22–126	142,254	0
Validation	127–153	36,548	312
Test	154–180	36,059	175

StandardScaler fitted on normal training data only.

5.2 Autoencoder

Architecture: $30 \rightarrow 20 \rightarrow 10 \rightarrow 20 \rightarrow 30$. Anomaly score: mean squared reconstruction error. Trained 50 epochs, batch size 256, Adam optimizer ($\text{lr} = 10^{-3}$). Best model selected by validation AUC.

	Previous	Redesigned
Validation AUC	—	0.9923
Test AUC	0.577	0.9935

Per-scenario test AUC:

Scenario	AUC	n
cheque_structuring	1.0000	31
smurfing	1.0000	35
round_amount	1.0000	2
amount_spike	0.9998	3
duplicate_virement	0.9980	2
frequency_burst	0.9983	48
repeated_client_employee	0.9983	16
timing_anomaly	0.9710	12
cumulative_threshold	0.9742	26

5.3 LSTM

Architecture: LSTM(input=30, hidden=64, layers=2) $\rightarrow$ Linear(64, 30). Sequence length: 10 events per client. Anomaly score: mean squared prediction error on the target event. Trained 50 epochs, batch size 256, Adam (lr = 10^{-3}). Best model at epoch 5.

	Previous	Redesigned
Validation AUC	—	0.9473
Test AUC	0.564	0.9430

Per-scenario test AUC:

Scenario	AUC	n
round_amount	1.0000	1
amount_spike	0.9999	1
cheque_structuring	0.9999	13
smurfing	0.9981	7
repeated_client_employee	0.9976	10
frequency_burst	0.9125	38
cumulative_threshold	0.8540	7

5.4 Observation: Difficulty Inversion

An unexpected finding: “hard” scenarios (AE: 0.9996, LSTM: 0.9987) scored higher than “medium” scenarios (AE: 0.9872, LSTM: 0.9034). This occurs because difficulty was defined from a *human analyst’s* perspective (smurfing is hard for a person to spot), while the model sees it differently — smurfing events produce extreme values across multiple features simultaneously (near-threshold amount + forced operation type shift + elevated near_threshold_count_7d), yielding high reconstruction/prediction error. Medium scenarios like timing_anomaly or cumulative_threshold produce subtler single-dimension shifts that are harder for the model to flag.

6 Architectural Decisions

ID	Decision
D23	Enriched event redesigned as contrast features (event vs profile), not profile dump. 30 features in 3 categories: raw event (7), contrast (17), context (6).
D24	Oracle profile baseline: client personality parameters exported directly from generator objects, replacing the weekly snapshot pipeline. Production note documented.
D25	Velocity features computed from per-client rolling buffer (size 50) at enrichment time, not from weekly profile snapshots.
D26	Anomaly scenarios reduced from 13 to 9. Five cut for scope/feasibility; repeated_client_employee added to preserve dual-actor model.
D27	Anomalous events injected additively alongside normal events (not in-place modification). Anomaly window checked before daily probability.

7 Saved Artifacts

File	Contents
models/autoencoder.pt	Autoencoder best weights
models/lstm.pt	LSTM best weights
models/scaler.pkl	Fitted StandardScaler (shared)
models/feature_cols.json	Ordered feature column list
models/lstm_config.json	LSTM architecture parameters
data/transactions.csv	244k raw events
data/clients.csv	10k client oracle profiles
data/enriched_events.csv	244k enriched events (30 features)

8 Next Steps

1. Score fusion strategy: combining Autoencoder + LSTM anomaly scores
2. Scoring Service implementation (model loading, SHAP integration)
3. Decision Service (alert tier assignment: INFO / REVIEW / ALERT / BLOCK)
4. Streaming pipeline (RedPanda topics: scored-events → decisions)
5. Dashboard (Streamlit) consuming decisions topic